Physics - 0   2026 Spring

Homework 2

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*All of my homework is available at `https://github.com/lacampanella233/Homework.git`.

Problem 1. (20 pts) Consider a Ferris wheel with radius R rotating at a constant angular velocity ω.

(a) (10pts) Find the weight you experience at the top and at the bottom of the Ferris wheel.

(b) (5pts) Find the angular velocity for which you would feel weightless.

(c) (5pts) To get some intuition for this number consider one of the largest Ferris wheels in the world, the Star of Nanchang (南昌之星) in Jiangxi province, which has a radius of $R = 80\,\mathrm{m}$. Assuming your weight is $70\,\mathrm{kg}$, what is the angular velocity and the period for which you would feel weightless? For comparison, the actual period of the Star of Nanchang is 30 minutes.

Solution. Suppose the person has mass m. Denote the support force experienced by the person at the top and the bottom of the Ferris wheel as N_{t} and N_{b} respectively. The weight of the person is mg.

(a) The person has an acceleration of $a = \omega^2 R$ towards the center of the Ferris wheel. At the top, we have

$$-N_{\mathrm{t}} + mg = m\omega^2 R \implies N_{\mathrm{t}} = mg - m\omega^2 R.$$

At the bottom, we have

$$N_{\mathrm{b}} - mg = m\omega^2 R \implies N_{\mathrm{b}} = mg + m\omega^2 R.$$

(b) To feel weightless, the support force at the top must be at most zero, so we have

$$mg - m\omega^2 R \leq 0 \implies \omega \geq \sqrt{\frac{g}{R}}.$$

(c) For $R = 80\,\mathrm{m}$, we have

$$\omega = \sqrt{\frac{g}{R}} = \sqrt{\frac{9.8\,\mathrm{m/s^2}}{80\,\mathrm{m}}} \approx 0.35\,\mathrm{rad/s}.$$

The period is

$$T = \frac{2\pi}{\omega} \approx \frac{2\pi}{0.35\,\mathrm{rad/s}} \approx 18\,\mathrm{s}.$$

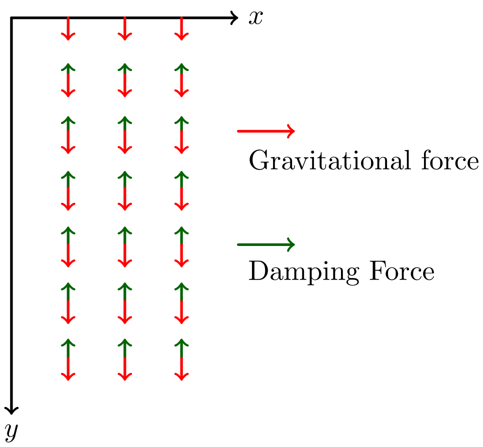
Problem 2. (25 pts) Consider a ball falling in some liquid. The liquid exerts a damping force whose magnitude is proportional to the squared speed of the ball, and whose direction is opposite to that of the velocity. Here is the damping constant. This force is known as quadratic drag and it holds for objects moving fast through viscous fluids. Suppose the ball is released from rest and has mass .

(a) (10pts) Draw the vector fields for the gravitational and damping forces.

(b) (10pts) Find the velocity as a function of time.

(c) (5pts) Determine the terminal velocity of the ball when .

Solution. (a) As shown below.



(b) Denote the velocity of the ball at time t as $v(t)$, with the positive direction being downward. We have

$$m\frac{\mathrm{d}v}{\mathrm{d}t} = mg - kv^2 \implies \frac{m\,\mathrm{d}v}{mg - kv^2} = \mathrm{d}t$$

$$\implies \int_0^{v(t)} \frac{m\,\mathrm{d}v}{mg - kv^2} = t \implies t = \sqrt{\frac{m}{gk}}\operatorname{arctanh}\left(\sqrt{\frac{k}{mg}}v(t)\right)$$

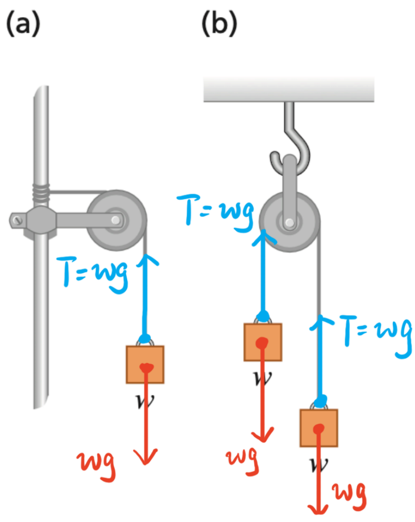
$$\implies v(t) = \sqrt{\frac{mg}{k}}\tanh\left(\sqrt{\frac{gk}{m}}t\right).$$

(c) Take $t \to +\infty$, we got the terminal velocity

$$v_{\text{terminal}} = \sqrt{\frac{mg}{k}}.$$

Problem 3. (20 pts) In the figure below, each of the blocks was weight w. Assuming the pulleys are frictionless and the ropes are massless, draw a diagram of the forces acting on the blocks in each case, and calculate the tension in the rope in terms of w.

Solution. As shown below.

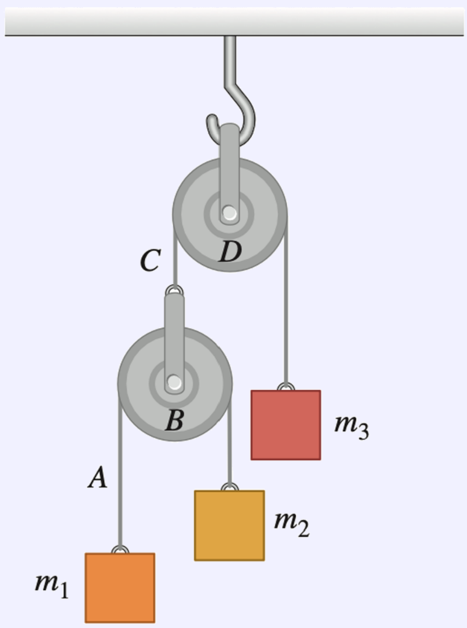


Problem 4. (35 pts) Consider the system illustrated in the figure below where the masses m_1, m_2, and m_3 are arbitrary. Let the system be released from rest. Assuming that the pulleys are frictionless and the ropes are massless, compute the following:

(a) (15 pts) The acceleration of each of the blocks.

(b) (10 pts) The tension in the rope A and C.

(c) (10 pts) What happens to answers (a) and (b) when $m_1 = m_2 = m$ and $m_3 = 2m$? Is this the expected results?



Solution. (a) and (b) Denote the distance from the ceiling to m_1, m_2, m_3 as x_1, x_2, x_3 respectively, and the acceleration of m_1, m_2, m_3 as a_1, a_2, a_3 respectively, with the positive direction being downward.

We immediately got the relation

$$a_i = \ddot{x}_i \ (1 \le i \le 3), \quad x_1 + x_2 + 2x_3 = \text{const.} \implies a_1 + a_2 + 2a_3 = 0. \tag{1}$$

Denote the tension in rope A and C as T_A and T_C respectively. Since the pulley B is massless, we have $2T_\mathrm{A} = T_\mathrm{C}$, hence we can assume that

$$T_\mathrm{A} := T, \quad T_\mathrm{C} := 2T. \tag{2}$$

Consider Newton's second law for each block, we have

$$\begin{cases} m_1 a_1 &= m_1 g - T_\mathrm{A}, \\ m_2 a_2 &= m_2 g - T_\mathrm{A}, \\ m_3 a_3 &= m_3 g - T_\mathrm{C}. \end{cases}$$

Combining with the relation (1) and (2) and solving those equations, we got

$$\begin{cases} a_1 = \dfrac{4m_1m_2 + m_1m_3 - 3m_2m_3}{4m_1m_2 + m_1m_3 + m_2m_3} g, \\ a_2 = \dfrac{4m_1m_2 - 3m_1m_3 + m_2m_3}{4m_1m_2 + m_1m_3 + m_2m_3} g, \\ a_3 = \dfrac{-4m_1m_2 + m_1m_3 + m_2m_3}{4m_1m_2 + m_1m_3 + m_2m_3} g, \\ T_\mathrm{A} = \dfrac{4m_1m_2m_3 g}{4m_1m_2 + m_1m_3 + m_2m_3}, \\ T_\mathrm{C} = \dfrac{8m_1m_2m_3 g}{4m_1m_2 + m_1m_3 + m_2m_3}. \end{cases}$$

(c) Take $m_1 = m_2 = m$ and $m_3 = 2m$, we have

$$\begin{cases} a_1 = a_2 = 0, \\ a_3 = 0, \\ T_\mathrm{A} = mg, \\ T_\mathrm{C} = 2mg. \end{cases}$$

This is the expected result, as the system is in equilibrium when the masses are given as above.